

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	261498	50.032813
1	[Git, Change, Create]	92826	17.760541
2	[Git, Change, Delete]	58957	11.280333
3	[Git, Change, Rename]	24301	4.649548
4	[File, Git]	16199	3.099380
5	[Git, Commit]	15955	3.052695
6	[Java, ByteCode, Parameter]	9236	1.767138
7	[Java, ByteCode, Member, Method]	8981	1.718349
8	[Java, ByteCode, Bound]	6321	1.209407
9	[Java, Value, ByteCode, Annotation]	5389	1.031086
10	[Java, ByteCode, ParameterizedType, Bound]	4895	0.936568
11	[Xml, Element]	2416	0.462257
12	[Java, ByteCode, Member, Field]	2208	0.422460
13	[Xml, Text]	1622	0.310340
14	[Java, ByteCode, Member, Constructor, Method]	1596	0.305365
15	[Java, ByteCode, Bound, WildcardType]	970	0.185592
16	[Java, ByteCode, Member, Method, Lambda]	851	0.162823
17	[Java, ByteCode, Bound, TypeVariable]	848	0.162249
18	[Type, File, Java, ByteCode, JavaType]	690	0.132019
19	[Type, File, Java, Class, ByteCode]	660	0.126279
20	[Json, Key]	544	0.104084
21	[Value, Json, Scalar]	534	0.102171
22	[Git, Change, Copy]	481	0.092030
23	[Type, File, Java, ByteCode, ResolvedDuplicate...]	477	0.091265
24	[Java, ByteCode, Member, Method, GenericDeclar...]	466	0.089160
25	[Java, Value, ByteCode, Primitive]	329	0.062948
26	[Type, File, Java, ByteCode, ExternalType]	313	0.059887
27	[Author, Git, Person]	304	0.058165
28	[Committer, Git, Person]	247	0.047259
29	[Maven, Dependency]	195	0.037310

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

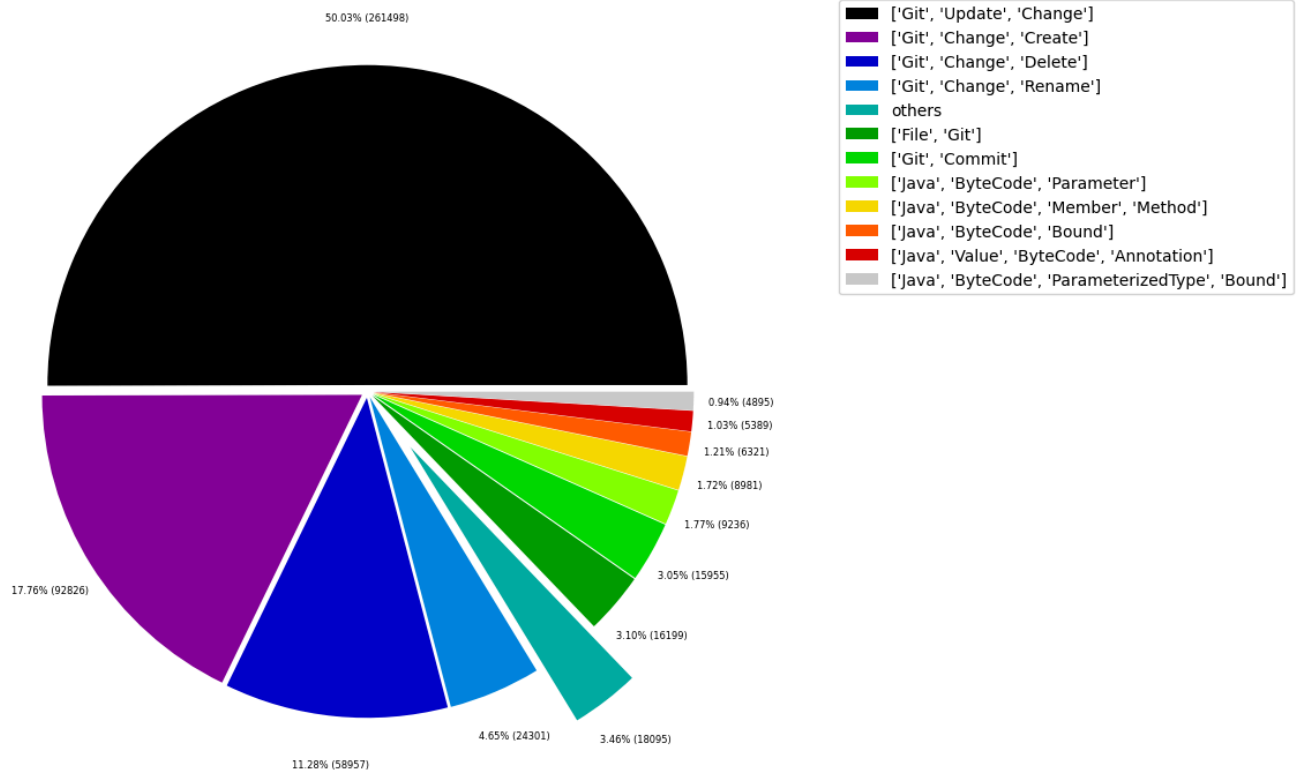


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000191
1	[Git, Branch]	1	0.000191
2	[Repository, File, Git]	1	0.000191
3	[File, Json]	2	0.000383
4	[File]	3	0.000574
5	[Maven, Exclusion]	4	0.000765
6	[Type, File, Java, ByteCode, Record, GenericDe...	4	0.000765
7	[Value, Array, Json]	8	0.001531
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001531
9	[Java, ByteCode, Member, Constructor, Method, ...	9	0.001722
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002105
11	[Maven, Scm]	11	0.002105
12	[File, Maven, Xml, Pom, Document]	11	0.002105
13	[Type, File, Java, ByteCode, Void]	11	0.002105
14	[Java, ManifestSection]	11	0.002105
15	[File, Java, Manifest]	11	0.002105
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002105
17	[File, Java, ServiceLoader]	12	0.002296
18	[Maven, PluginExecution]	13	0.002487
19	[File, Java, Properties]	13	0.002487
20	[Maven, ExecutionGoal]	13	0.002487
21	[Type, File, Java, ByteCode, Enum]	17	0.003253
22	[jQAssistant, Rule, Concept]	19	0.003635
23	[Maven, Configuration]	20	0.003827
24	[Maven, Plugin]	20	0.003827
25	[Xml, Attribute]	22	0.004209
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007462
27	[Type, File, Java, ByteCode, Annotation]	42	0.008036
28	[Type, File, Java, Class, ByteCode, Throwable]	42	0.008036
29	[Xml, Namespace]	44	0.008419

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

	['Xml', 'Element']
	['Java', 'ByteCode', 'Member', 'Field']
	['Xml', 'Text']
	['Java', 'ByteCode', 'Member', 'Constructor', 'Method']
	['Java', 'ByteCode', 'Bound', 'WildcardType']
	['Java', 'ByteCode', 'Member', 'Method', 'Lambda']
	['Java', 'ByteCode', 'Bound', 'TypeVariable']
	['Type', 'File', 'Java', 'ByteCode', 'JavaType']
	['Type', 'File', 'Java', 'Class', 'ByteCode']
	['Json', 'Key']
	['Value', 'Json', 'Scalar']
	others
	['Git', 'Change', 'Copy']
	['Type', 'File', 'Java', 'ByteCode', 'ResolvedDuplicateType']
	['Java', 'ByteCode', 'Member', 'Method', 'GenericDeclaration']
	['Java', 'Value', 'ByteCode', 'Primitive']
	['Type', 'File', 'Java', 'ByteCode', 'ExternalType']
	['Author', 'Git', 'Person']
	['Committer', 'Git', 'Person']
	['Maven', 'Dependency']
	['Type', 'File', 'Java', 'ByteCode', 'Interface']
	['Value', 'Array']
	['Package', 'File', 'Directory', 'Java']
	['Git', 'Tag']
	['Java', 'Value', 'ByteCode', 'Enum']
	['Value', 'Json', 'Object']
	['Value', 'Property']
	['Type', 'File', 'Java', 'Class', 'ByteCode', 'GenericDeclaration']
	['Java', 'ByteCode', 'Bound', 'GenericArrayType']
	['Type', 'File', 'Java', 'ByteCode', 'Throwable', 'JavaType']
	['Artifact', 'Maven']
	['Value', 'ManifestEntry']
	['Java', 'Value', 'Class', 'ByteCode']
	['Type', 'File', 'Java', 'ByteCode', 'Interface', 'GenericDeclaration']
	['Type', 'File', 'Java', 'ByteCode', 'ExternalType', 'ExternalAnnotation']
	['File', 'Directory']

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

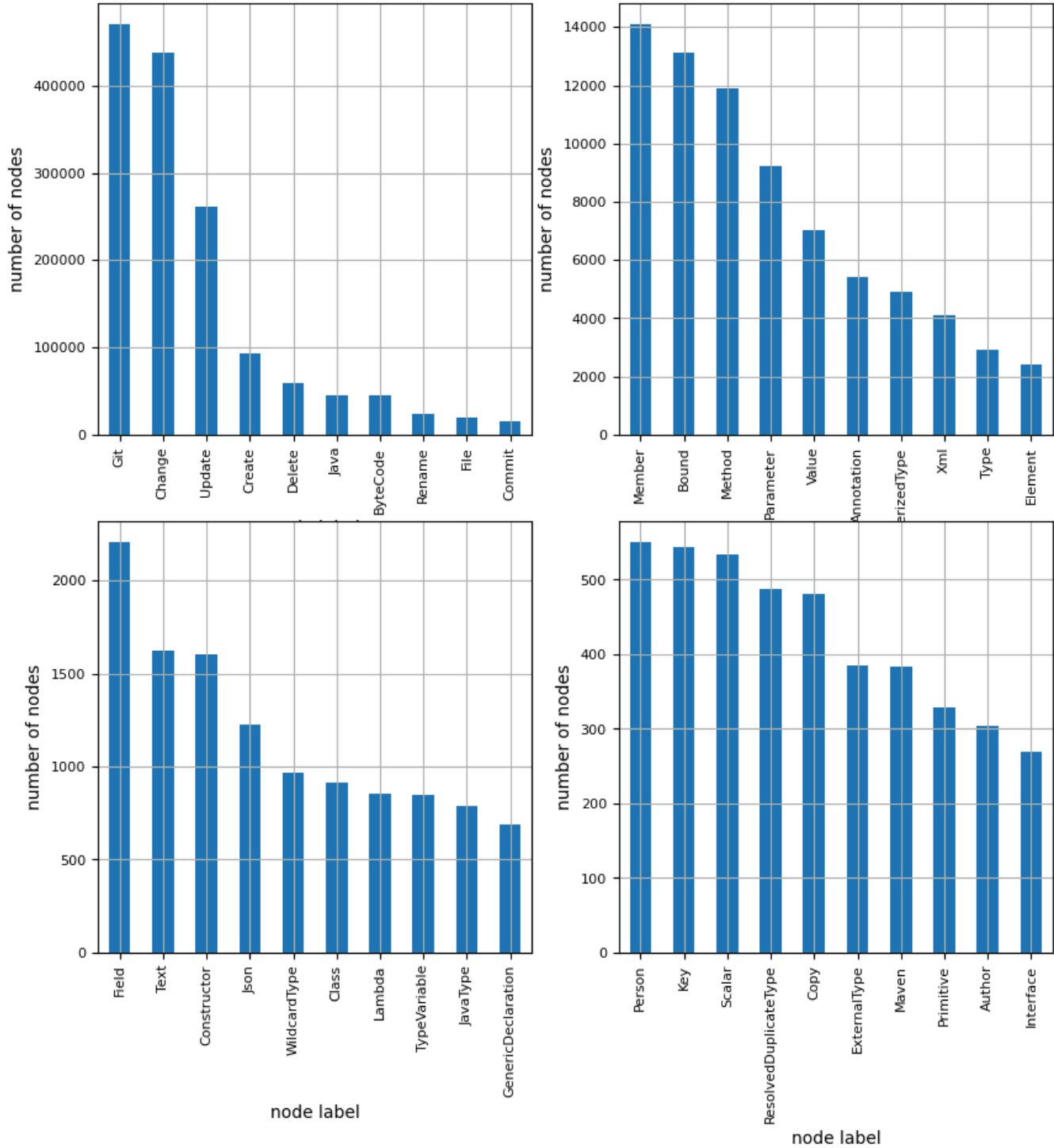
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	470915	90.100889
1	Change	438063	83.815266
2	Update	261498	50.032813
3	Create	92826	17.760541
4	Delete	58957	11.280333
5	Java	45554	8.715917
6	ByteCode	45350	8.676885
7	Rename	24301	4.649548
8	File	19387	3.709344
9	Commit	15955	3.052695
10	Member	14111	2.699879
11	Bound	13137	2.513522
12	Method	11903	2.277419
13	Parameter	9236	1.767138
14	Value	7016	1.342382
15	Annotation	5431	1.039122
16	ParameterizedType	4895	0.936568
17	Xml	4115	0.787329
18	Type	2923	0.559262
19	Element	2416	0.462257
20	Field	2208	0.422460
21	Text	1622	0.310340
22	Constructor	1605	0.307087
23	Json	1223	0.233998
24	WildcardType	970	0.185592
25	Class	912	0.174494
26	Lambda	851	0.162823
27	TypeVariable	848	0.162249
28	JavaType	789	0.150961
29	GenericDeclaration	686	0.131253
30	Person	551	0.105424
31	Key	544	0.104084
32	Scalar	534	0.102171
33	ResolvedDuplicateType	488	0.093370
34	Copy	481	0.092030
35	ExternalType	385	0.073663
36	Maven	383	0.073280
37	Primitive	329	0.062948
38	Author	304	0.058165
39	Interface	269	0.051468

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1622122

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	438063	27.005552
1	MODIFIES	438063	27.005552
2	UPDATES	261498	16.120736
3	CREATES	117608	7.250256
4	DELETES	83258	5.132660
5	COMMITTED	31910	1.967176
6	RENAMES	24301	1.498099
7	INVOKES	22722	1.400758
8	HAS_PARENT	19403	1.196149
9	OF_TYPE	18564	1.144427
10	DEPENDS_ON	17816	1.098314
11	HAS_FILE	16199	0.998630
12	HAS_COMMIT	15955	0.983588
13	DECLARES	14459	0.891363
14	OF_RAW_TYPE	13189	0.813071
15	HAS	9866	0.608216
16	HAS_NEW_NAME	9446	0.582324
17	RETURNS	8167	0.503476
18	READS	6081	0.374879
19	HAS_ACTUAL_TYPE_ARGUMENT	5811	0.358234
20	ANNOTATED_BY	5388	0.332158
21	OF_GENERIC_TYPE	4376	0.269770
22	CHANGED_TOGETHER_WITH	4036	0.248810
23	SIMILAR	3419	0.210773
24	CONTAINS	3227	0.198937
25	RESOLVES_TO	2683	0.165401
26	RETURNS_GENERIC	2601	0.160346
27	HAS_FIRST_CHILD	2416	0.148941
28	HAS_LAST_CHILD	2416	0.148941
29	HAS_ELEMENT	2394	0.147584

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

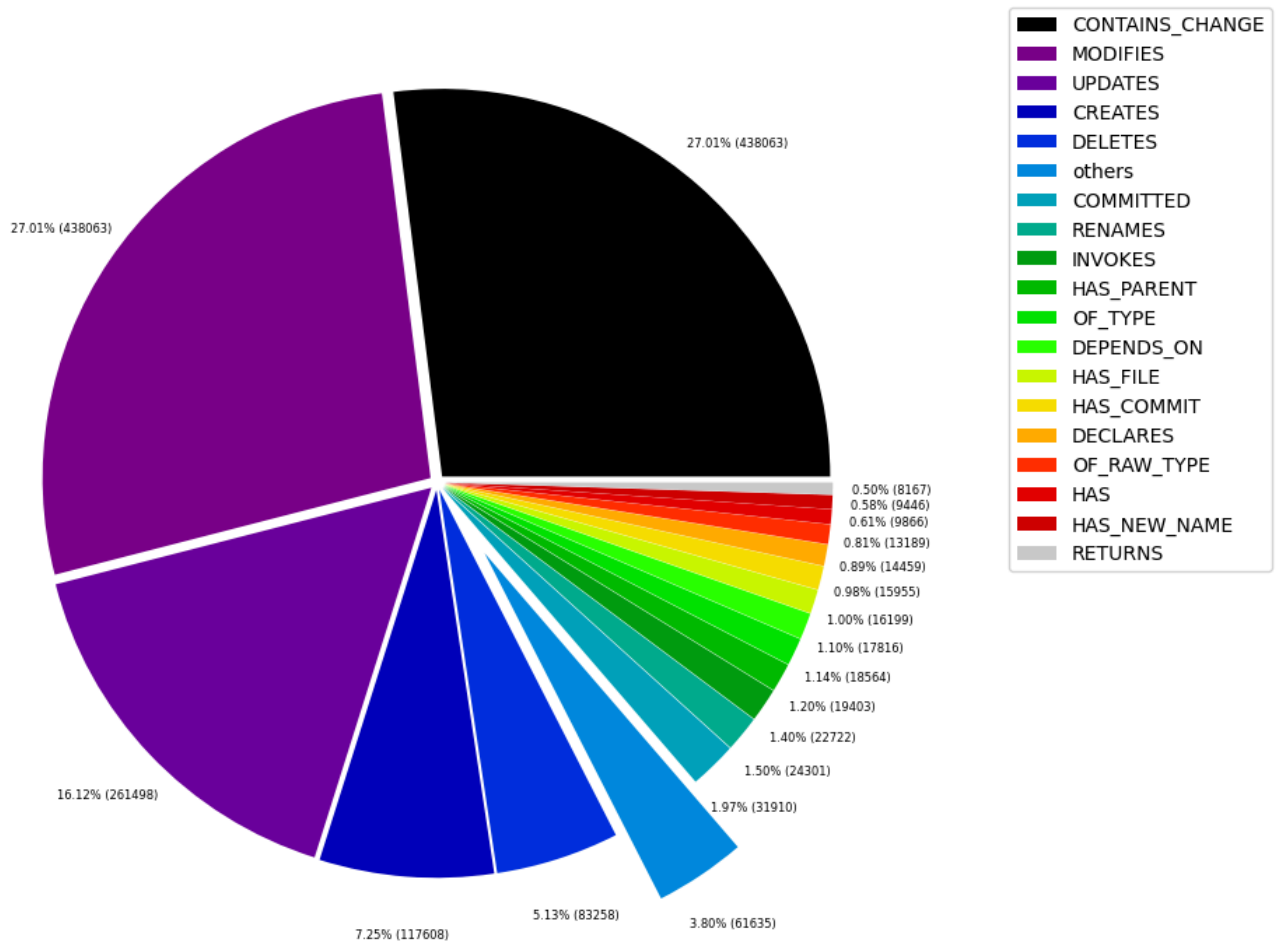


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

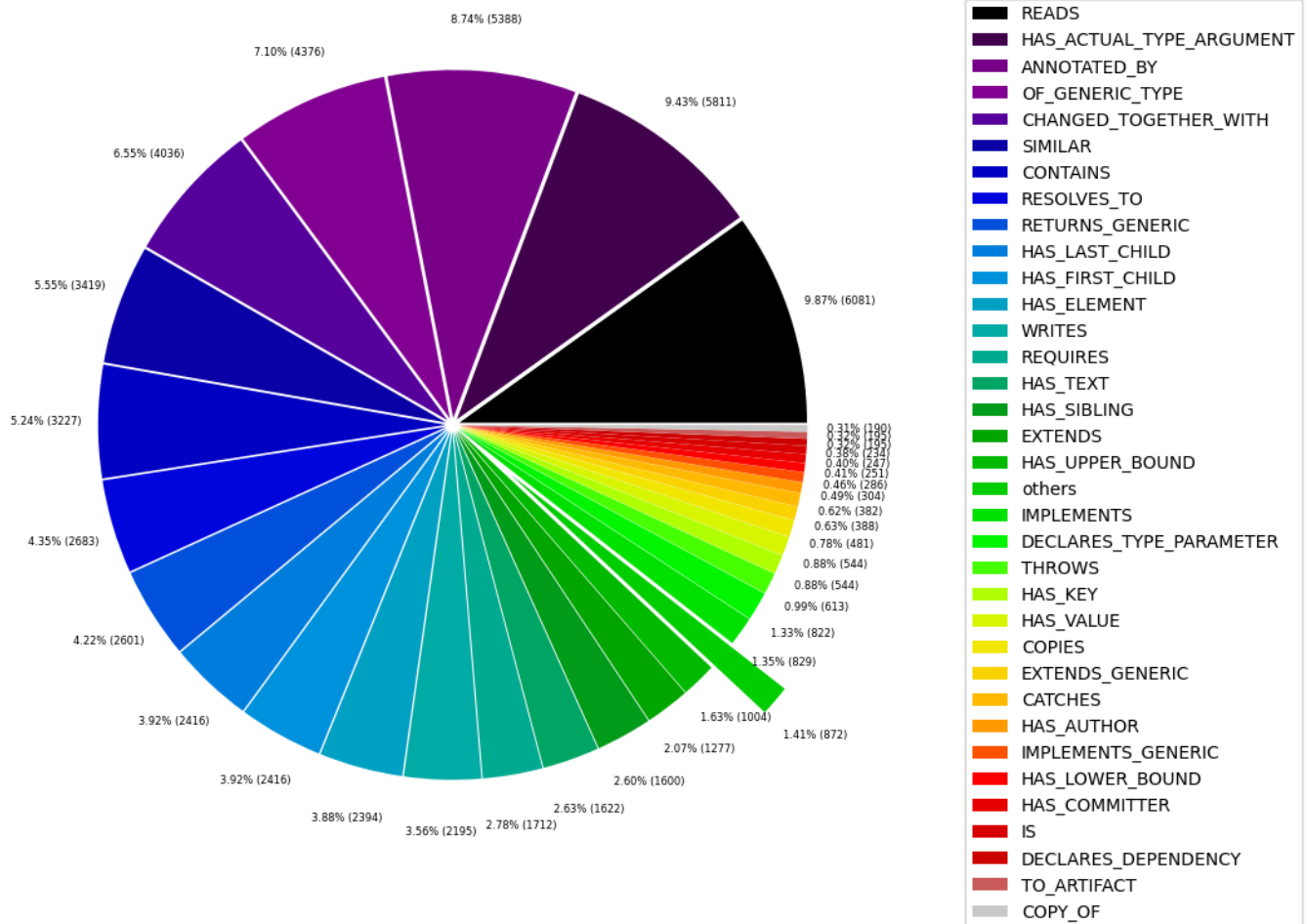
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000062
1	HAS_BRANCH	1	0.000062
2	HAS_HEAD	2	0.000123
3	EXCLUDES	4	0.000247
4	THROWS_GENERIC	5	0.000308
5	HAS_SCM	11	0.000678
6	DESCRIBES	11	0.000678
7	HAS_ROOT_ELEMENT	13	0.000801
8	HAS_GOAL	13	0.000801
9	HAS_EXECUTION	13	0.000801
10	INCLUDES_CONCEPT	19	0.001171
11	IS_ARTIFACT	20	0.001233
12	HAS_CONFIGURATION	20	0.001233
13	USES_PLUGIN	20	0.001233
14	OF_NAMESPACE	22	0.001356
15	HAS_ATTRIBUTE	22	0.001356
16	REQUIRES_TYPE_PARAMETER	26	0.001603
17	REQUIRES_CONCEPT	28	0.001726
18	DECLARES_NAMESPACE	44	0.002712
19	HAS_DEFAULT	53	0.003267
20	HAS_COMPONENT_TYPE	103	0.006350
21	CONTAINS_VALUE	131	0.008076
22	HAS_TAG	145	0.008939
23	ON_COMMIT	145	0.008939
24	COPY_OF	190	0.011713
25	TO_ARTIFACT	195	0.012021
26	DECLARES_DEPENDENCY	195	0.012021
27	IS	234	0.014426
28	HAS_COMMITTER	247	0.015227
29	HAS_LOWER_BOUND	251	0.015474

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodes
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	261498	15955	
1	[Git, Update, Change]	MODIFIES	[File, Git]	261498	261498	
2	[Git, Update, Change]	UPDATES	[File, Git]	261498	261498	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	92826	15955	
4	[Git, Change, Create]	CREATES	[File, Git]	92826	92826	
5	[Git, Change, Create]	MODIFIES	[File, Git]	92826	92826	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	58957	15955	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	58957	58957	
8	[Git, Change, Delete]	DELETES	[File, Git]	58957	58957	
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24301	15955	
10	[Git, Change, Rename]	DELETES	[File, Git]	24301	24301	
11	[Git, Change, Rename]	CREATES	[File, Git]	24301	24301	
12	[Git, Change, Rename]	MODIFIES	[File, Git]	24301	24301	
13	[Git, Change, Rename]	RENAMES	[File, Git]	24301	24301	
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19392	15955	
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16199	1	
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15955	1	
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	15955	304	
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15955	247	
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13614	8951	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9446	16199	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5451	8951	
22	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5331	8951	
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalType, Ext...	5035	5389	
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	4123	9236	
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3821	9236	
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3341	16199	
27	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2826	4895	
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2394	2416	
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2297	6321	

Graph Density

total_number_of_nodes (vertices): 522653

total_number_of_relationships (edges): 1622122

-> total directed graph density: 5.9382359553048965e-06

-> total directed graph density in percent: 0.0005938235955304897